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Modeling and implementing an earthquake and tsunami event-triggered, time-constrained impact assessment workflow

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Abstract

Disaster warning and emergency response are an emerging field of HPC/cloud computing which is now known as *urgent computing*. This means that computing is required to be performed within short time scales. In this regard, the timeliness for early warning of the population is the foremost important point well before accuracy, as long as the predictions are sufficiently accurate, hence a coarser level of precision is acceptable to that end. It also processes a large amount of data and provides well-targeted, high-resolution, and highly reliable information to emergency management stakeholders. At an early stage, it is also important to take into account the uncertainties surrounding the disaster parameters. To do this, the simulation process must include a range of possible values for each of the main input parameters. Following this principle, we hereby present an impact assessment workflow that is designed to provide building-specific damage assessments caused by earthquakes and possible subsequent tsunamis. We target short run-times to guarantee that results are provided as needed for emergency response decisions and that a proper impact assessment is provided, allowing for meaningful planning of rescue actions. We show the design principles coming from a time-aware model of computation, to the specification of the workflow, and the individual programs that compose it. A working prototype was developed as part of the LEXIS project. This paper also includes preliminary experiments toward robust urgent computing, utilizing distributed, heterogeneous HPC and cloud infrastructures.

Keywords

Models of computation, time-constrained scheduling, emergency response, tsunami simulation, SEM

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1. Introduction

Since its inception in the mid-1950s, civilian utilization of High Performance Computing (HPC) has spread to ever-growing fields of applications. Nowadays, the cloud infrastructure has become pervasive either by the use of large scale provider (e.g. AWS, Google, Microsoft, …) or more local providers. At the same time, the scope of applications went from engineering and prototyping to more mundane and day-to-day but high-impact ones. The newest fields include personal healthcare and medicine, traffic monitoring, and automotive float monitoring. However, on the horizon, the main field of applications leans toward digital twins: utilizing supercomputers to simulate the outcome of a given operation to something before applying it to the real world, or (as this is the focus of this paper) conversely predicting the outcome of a given event in the real world before any large scale observation can be made on it.

With the advent of a significant processing power at a relatively low cost, it is now theoretically possible to predict some of the outcomes of given natural disaster. The associated term with this field of application is "urgent computing". It allows one to anticipate bad outcomes of an event, or act efficiently with emergency response teams when an overwhelming event can disrupt local infrastructures. This is a significant expected outcome of our study since the best-allocated resources can avoid further deterioration of the situation when a disaster occurs.

This paper presents a working prototype of this use-case of urgent computing applied to an earthquake event, including potential tsunami-related inundations both before the occurrence of most of the consequences and at an early stage after the outcomes. This workflow was aimed at being one of the essential study cases for the LEXIS platform developed in the eponymous project –Golasowski et al. (2022)– and drove the development of several tools part of the platform to enable this urgent computing demonstrator.

Our main contribution is the case for a workflow able to run on several sites potentially with redundancies and able to enforce time constraints and a level of timeliness. Both redundancy and timeliness are important properties to enable the case for urgent computing over cloud and HPC infrastructures. To that end, the key factor in the design is a time-aware model of computation (MoC). Its implementation was possible thanks to several novel components of the LEXIS infrastructure and architecture. These include, in particular, the distributed data infrastructure (DDI) filesystem and the York orchestrator, both already described in Hachinger et al. (2022).

In this paper, we focus on the base of a real-time-aware MoC and how it drives the design of the urgent computing workflow, then and we show the main components of the workflow: TsunAWI, the tsunami simulation tool, the earthquake loss assessment, and the fusion of the data in a Satellite-based Emergency Mapping.

2. Models of computation and design

The timeliness of an earthquake and tsunami damage assessment for emergency planning requires a special design, comparable in some way to what can be found in embedded real-time (safety critical) applications observed in e.g. aeronautics software requirements. Whilst the time scales of embedded applications and HPC/cloud based damage assessment are not comparable (typically a few mili-seconds for the former vs a few dozens of seconds or minutes for the latter), ensuring time constraints within a sound design draws a lot of similarities.

Indeed, the desirable properties would be in each case to have:

- A sound parallelism design and implementation within each software component;
- A level of redundancy that would permit some output even in the case of downtime on e.g. an HPC center or a cloud provider;
- A level of guaranteed time-constraints from the event to the Emergency Mapping.

To this end, we relied on a model of Computation (MoC), whose principles are described in this section. It is inherited from the embedded software design domain. While not all the desired properties listed above were guaranteed in our initial working workflow due to limited integration time, each individual component successfully passed validation tests for their respective portions of the workflow.

2.1. Model of computation (MoC)

A model of computation (MoC) is a formalization of how a computing system works. For single-core computers, an early model is the well known Von Neumann model in which a processor fetches instructions and works on data from memory. In the early days of computer architecture and programming several MoCs were defined, e.g. stack based computing which provided the theoretical underlying base for the Forth language, whereas the Von Neumann model provides the foundation for all sorts of imperative programming such as Fortran and C, later enhanced with the Object Oriented paradigms.

For distributed computing, several models were introduced from the 1970s and later. An important example is Kahn Process Network as defined by Kahn (1974) (KPN) which provides a formalization of the utilization of processes and pipes (or communication channels).

KPN instances define processes (i.e. communicating programs/components) with communication channels whose quantum of communication is called a token (an

element of data produced and consumed atomically). A channel connects a process output port to another process input port. The topology is fixed, thus defining a graph whose nodes are processes (the P in KPN) and the edges are the communication channels transporting data tokens. Each process can have input channels and want to read and consume input data tokens on them. If any input channel has not the requested number of data tokens, then the process is stalled until that number of tokens is at least reached. This behavior defines an implicit scheduling of the executions. But as much as the KPN model is useful to design distributed systems, it does not provide any guarantees of liveness or determinism. That is why others MoCs were derived to that end.

One of the successful model of computation for embedded distributed computing is Synchronous Data Flow (SDF) defined by Lee and Messerschmitt (1987) and its derivatives as e.g. Cyclo-Static Data-Flow (CSDF Bilsen et al. (1996)) for which the verification of consistency[1] and determinism can be done at the design phase with a polynomial algorithm. For real-time systems, there are all sorts of MoCs among which the Synchronous Models (not to be mistaken with SDF) are popular because of their tools and language support like Esterel –Berry (2000)– and their successes in modeling very low latency real-time systems. Nonetheless when the system is distributed and the latencies can not be neglected, some extensions of Time-aware SDF and CSDF are proven to work in a reliable way for real-time applications and system design of time-dependent, deterministic applications.

The determinism within a MoC means that any same input is guaranteed to give the same output independently of discrepancies in scheduling, communication timing, process fine-grain firing, or allocation of computing resources. Therefore, it is a desirable property of a MoC since that it allows to replay error scenarios, debug a given workflow by replaying an error scenario previously encountered, or more powerfully enabling validation of the whole system by validating and testing all its parts.

Another important property of a MoC is its ability of abstracting the design and implementation of a given complex specification in a easy to grasp and verify model. Below, we explain the bases of the MoC we utilize and showcase how it abstract the design in a very simple way.

2.2. Bases of static data-flow graphs

A first simple approach to data-flow graphs is to define them in a simple form as a triple (P, C, T) where P is a set of processes $P = \{p_1, \ldots, p_m\}$, $C = \{c_1, \ldots, c_n\}$ is a set of communication channels, and finally $T = \{t_1, \cdots t_n\}$ a set of data-tokens associated with each communication edge in C. P composes the nodes of the graph. C corresponds to the edges of the graph $C \subset P \times P$.

To take into account that data-tokens are specified with a given type (e.g. an image, a integer, a floating point value, etc.), one needs to define typed input and output communication ports for each process. Then, any communication channel $c_\nu \in C$ join an output port $o_{\mu,\nu}$ of process p_μ to a given input port $i_{\mu',\nu}$. A prerequisite includes the fact that data-token types must match between an output port of a process, the communication channel and the input port of the next process. As the reader may notice, there is a single association between any given output communication port $o_{\mu,\nu}$, a communication channel c_ν and an input port of another process $i_{\mu',\nu}$, therefore, to simplify the notation we can use this association to assimilate a given channel and the corresponding output and input ports of the two processes it ties.

Static data-flow graphs consume data-token in their input channels at a fixed (constant) rate r_ν, by definition. If the number of tokens t_ν on channel c_ν is below that threshold ($t_\nu < r_\nu$) the corresponding process with input port i_ν is stalled. This modus operandi provides the requirements in term of synchronization, resource management and, in some extent, scheduling.

A simple instance of a SDF graph is shown in Figure 1. In this simple graph, each of the 3 processes has one input and one output port, therefore, there are 3 communication channels c_1 from P_1 to P_2, c_2 from P_2 to P_3 and c_3 from P_3 to P_1[2]. Associated to any SDF graph, one can define the topology matrix which represents the number of intake (negative number) or output (positive number) of data-tokens on the communication channels (rows) by the processes (columns). For the SDF graph of Figure 1, it would be:

$$\Gamma = \begin{pmatrix} 2 & -2 & 0 \\ 0 & 2 & -1 \\ -4 & 0 & 2 \end{pmatrix} \tag{1}$$

The first criterion for the SDF graph consistency would be to have a non-trivial left-nullspace of (1) which represents the number of activation of each process to return to the initial configuration of the graph in term of data-tokens on the channels. For our example, it is easy to find out that the vector $S = (1 \quad 1 \quad 2)$ is a possible solution, which means that by runing processes P_1 and P_2 one time, and P_3 twice, the graph returns to its initial configuration. Therefore the graph can run in bounded memory (with a limited capacity in the communication channels). The second condition for consistency is that, at least, one non-trivial set of firing according to vector Γ can be done. This is where the requirement of the initial 2 data-tokens in c_2 comes from: without this, as the graph is a loop, there is no way to run any process because they are all waiting for a number of tokens on their input port. This would stall the execution. Initializing c_2 with 2 data-tokens allows to run twice P_3 then once P_1 and finally P_2 once. Whilst in this example,

voluntarily simple, the execution of the process would be sequential, this would not be the case on a more elaborate one, and more importantly, it would not be a desired outcome. One can remark that it is easy to break the sequential execution by introducing more tokens (it could be e.g. 4 in c_2, but keeping 2 in c_2, it can also be 4 in c_3 as another possibility).

An often cited extension of SDF graphs is Cyclo-Static Data-Flow (CSDF) Bilsen et al. (1996). The main difference between SDF and SCDF is that for the former, the rate of consumption of data-tokens (r_v) is constant, whereas for the latter, the rate is a cycle of a fixed number (p_v) of constant rates $\rho_1, \ldots, \rho_{p_v}$ and we can define $r_v = \sum_{\alpha=1}^{p_v} \rho_\alpha$. CSDF allows for an easy formalization of data parallelism and processing parallelism without changing significantly the complexity of expressing compile-time guarantees on the liveness, lack of deadlocks, and most of the intrinsic issues generally associated to the management of parallelism.

An illustration of a simple CSDF graph is shown in Figure 2 and simply expands the previous one. Γ remains the same matrix as in (1) but for the scheduling, one has to multiply the number of executions (firings) of each process by the length of the rate sequence: $R_{P_1,c_1} = [1,0,1]$ so $|R_{P_1}| = 3$; the others are 2 and 1. The basic scheduling vector is therefore

$$S_{\text{csdf}} = (3 \quad 2 \quad 2)$$

CSDF also enable to define sub-sampling which is useful when clock rates are enforced. Nonetheless, for sub-sampling, other derived MoCs like Polygraph –Dubrulle et al. (2019)– are better fit.

2.2.1. Real-time and data-flow models. Enforcement of time-constraints in data-flow MoCs is a highly relevant topic for real-time and near real-time applications (also known as

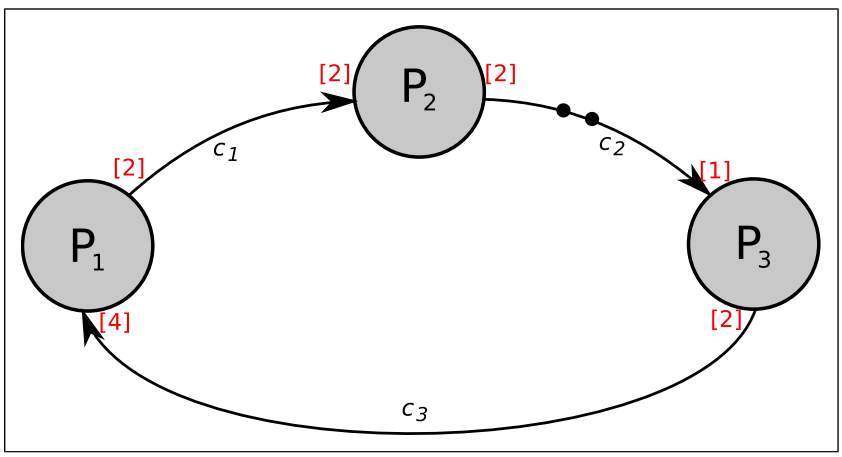

Figure 1. A simple example of a SDF graph with 3 processes $\{P_1, P_2, P_3\}$, 3 communication channels $\{c_1, c_2, c_3\}$ and 2 initial data-tokens in c_2. Each process only has 1 input port and 1 output port. Consumption/production rates on (resp.). Input/output ports are noted in square brackets. This SDF graph is self-consistent and the 2 initial tokens allows for this consistency and liveness.

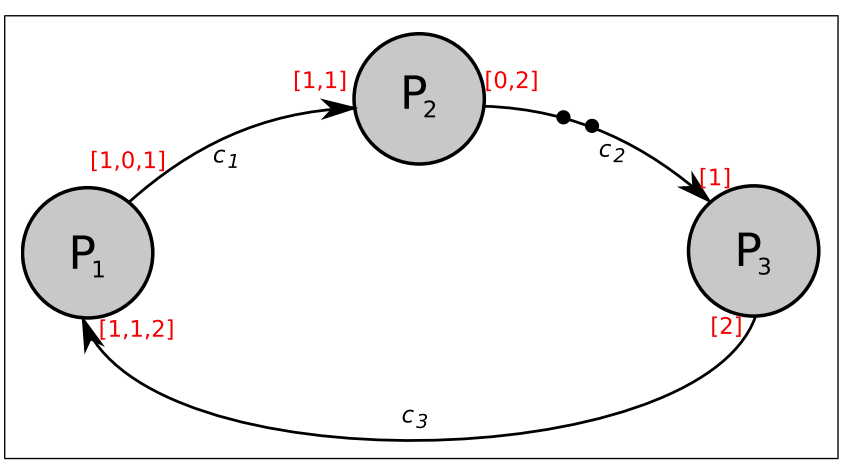

Figure 2. A simple example of a CSDF graph that expands the SDF graph from Figure 1.

"mixed critical" applications). We first need to be clear about the definition of a real-time system: a real-time system is a computing system with usually input and output related to the real-world and whose correctness is defined both by the correct output of the computing part but also by its timeliness. That is to say the temporal window formed by the delay between a given input from the real-world to the outcome of the computing is strictly specified and enforced. It can be several minutes and still be a real-time system, as long as it is specified this way. To be more specific, real-time is not necessarily about speed of processing, but about time determinism (even though the former is often a requirement to enforce the latter).

When implementing time and real-time constraints in a data-flow model, it is required to identify key processes that could prevent the timeliness of the application or in our case, the workflow. To that end, one is required to identify the conditions upon which a process within the workflow can be delayed: availability of computing resources (in embedded systems, this is especially tied to scheduling) on one hand, and the timely response of any fired process: from the beginning to the end of the execution of the process, the time window must be enforced. Data-flow models for embedded systems typically utilize a combination of formal time-constraint aware data-flow, especially deadline-aware-dataflows, e.g. Louise et al. (2014). Those can be implemented with watchdogs and specialized operating systems that include deadline aware scheduling. This is a new set of challenges for HPC centers and cloud computing because the availability of the computing resources in a timely fashion is not a granted property nor the capability to carry on computing as a deadline is triggered.

As shown in the next section, some elements of solution have been tested for both these issues, thanks to key features of the LEXIS platform. But first, we need to introduce further extensions for the MoC.

2.3. Data-centered processes

Extension of SDF and CSDF, encompassing derivatives such as Polygraph, can involve introducing data distribution and specific time-aware semantics that facilitate extended static analyses and abstract interpretation. This can be extended to graph transformation and optimization. The most commonly encountered ones include[3]:

- Duplicator (noted D_n): are processes with one input channel and n output channels such that any input data token on the input channel is copied on all the n output channels.
- Splitters (noted S_n): are processes with one input channel and n output channels that operate cyclically. Each time a new data token arrives on the input channel, it is distributed to an output channel which is incremented for the next input. This allows for a straightforward implementation of data parallelism[2].
- Joiners (noted J_n): are processes with n input channels and one output channel that increment their input channels and output the data tokens found on the input channel. Split-Join constructions allow for the typical *ForAll* kind of parallelism and preserve the order of the input channels.
- T-Joiners (noted J_n^T): are time-aware Joiners that, for the sake of timeliness, does not necessarily preserve the original order of the processing of data tokens. It is time triggered by the upstream Duplicator D_n or Splitter S_n so that the latency between the first output of upstream and the T-joiner is specified (deadline or latency aware) and usually strictly enforced (i.e. tokens produced after the deadline are discarded). It is worth noting that the output is a single data-token which aggregates all the values obtained in the $S_n \rightarrow J_n^T$ or $D_n \rightarrow J_n^T$ constructions i.e. within the time constraints. The way the individual data-elements are gathered in the output data token is usually by time of acquisition by the T-Joiner process, although there are other possibilities that are not relevant for this paper.

2.4. Application to the urgent-computing workflow

The actual implementation of the Earthquake and Tsunami workflow is illustrated in Figure 3 and its base data-flow model is shown in Figure 4.

There exists another parallel workflow that only starts after all the characteristics of the earthquake are known, with full duration and all the physical characteristics, so not only magnitude and coordinates as in the early warning workflow. In this later assessment workflow, the aim is rather to be a digital-twin of the affected area with high-resolution evaluations that can empower rescue teams and political teams to take informed decisions (SEM, Satellite-based Emergency Mapping). Outside of the fact that there is only one high-resolution TsunAWI process, and no enforced time-constraints, the associated workflow and MoC are very similar. The output is a high resolution map highlighting the areas of highest interest. This will be discussed thoroughly in the section about SEM.

3. A time-constrained workflow

This section describes the workflow that we designed using the time-constraint aware MoC as a support. It also discusses the implementation and tests done for ensuring the desired properties. As a rule in this paper, fully integrated parts are described on the reference (full-size) scenario of the Coquimbo (Chile) earthquake whilst some not yet integrated elements may be discussed on reduced size events.

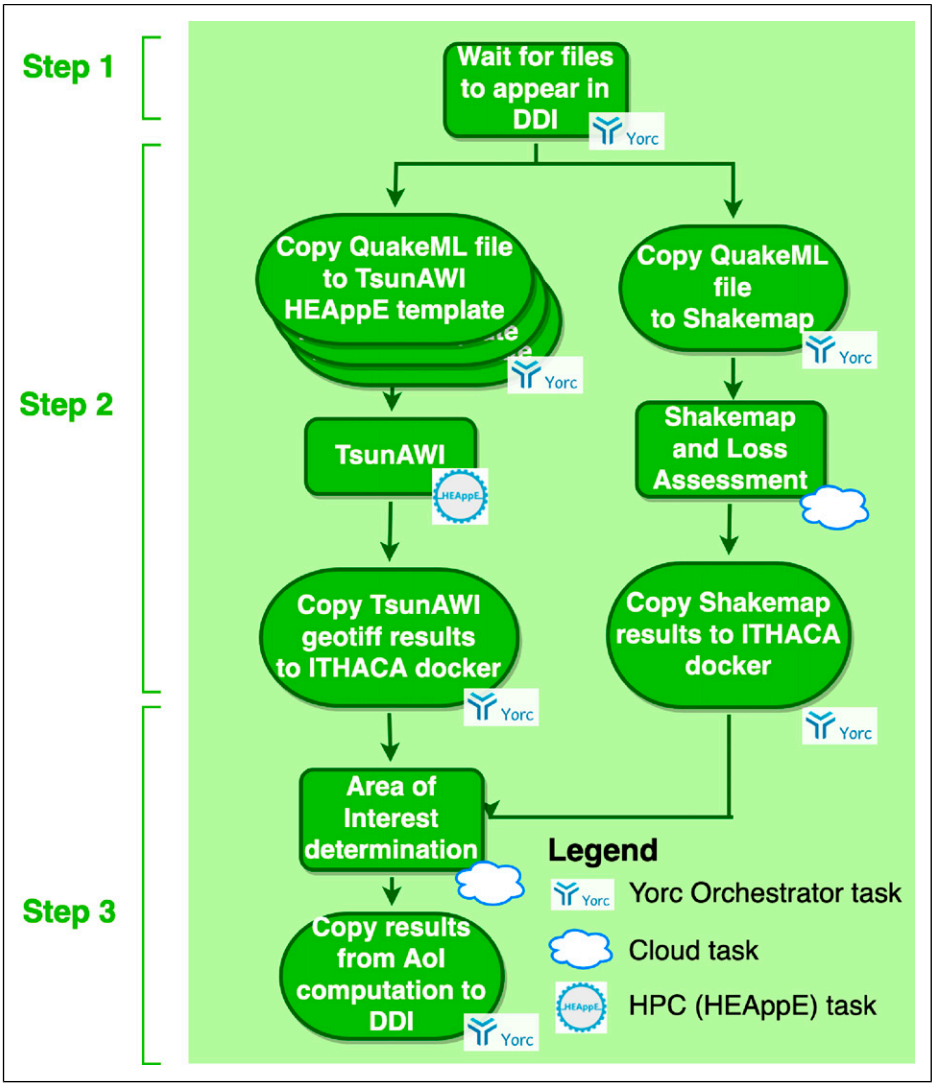

Figure 3. The complete earthquake and tsunami workflow overlaid over the LEXIS orchestration and data infrastructure. The relation with the MoC can be shown as follows: A specialized process waits for the the earthquake event (happenstance of a quakeml file on the DDI infrastructure of LEXIS as developed by ATOS), then, two data-paths are initiated, the loss assessment from the earthquake on one hand and an ensemble of TsunAWI fast (coarse-grain) simulations one the other hand. Both are fused as areas of interest. The final output is transmitted to relevant emergency management entities.

3.1. A target workflow on a specialized infrastructure

The LEXIS platform Hachinger et al. (2022) consists of multiple specialized components from which the most relevant for this publication are Yorc orchestrator ATOS-BULL (2020), HPC execution middleware HEAppE Svaton et al. (2020), and LEXIS DDI Munke et al. (2022). The Yorc orchestrator is responsible for the actual execution of the workflow as described. It deploys docker images on the cloud infrastructure and run them at appropriate time, in context of LEXIS platform, it uses LEXIS DDI for the data transfers, and it uses HEAppE middleware to execute the HPC tasks of the workflow. For interested readers the following book contains multiple articles describing individual parts of LEXIS platform in detail: Terzo and Martinovič (2022).

3.2. The demonstration workflow

As seen on Figure 3, what was demonstrated was both a specialized workflow taking advantage of the variety of features offered by the LEXIS platform, but also a first workflow that provides some elements of time-constraints.

Figure 3 is divided in 3 steps, and only the first one is required to run at any time: it acts as an earthquake

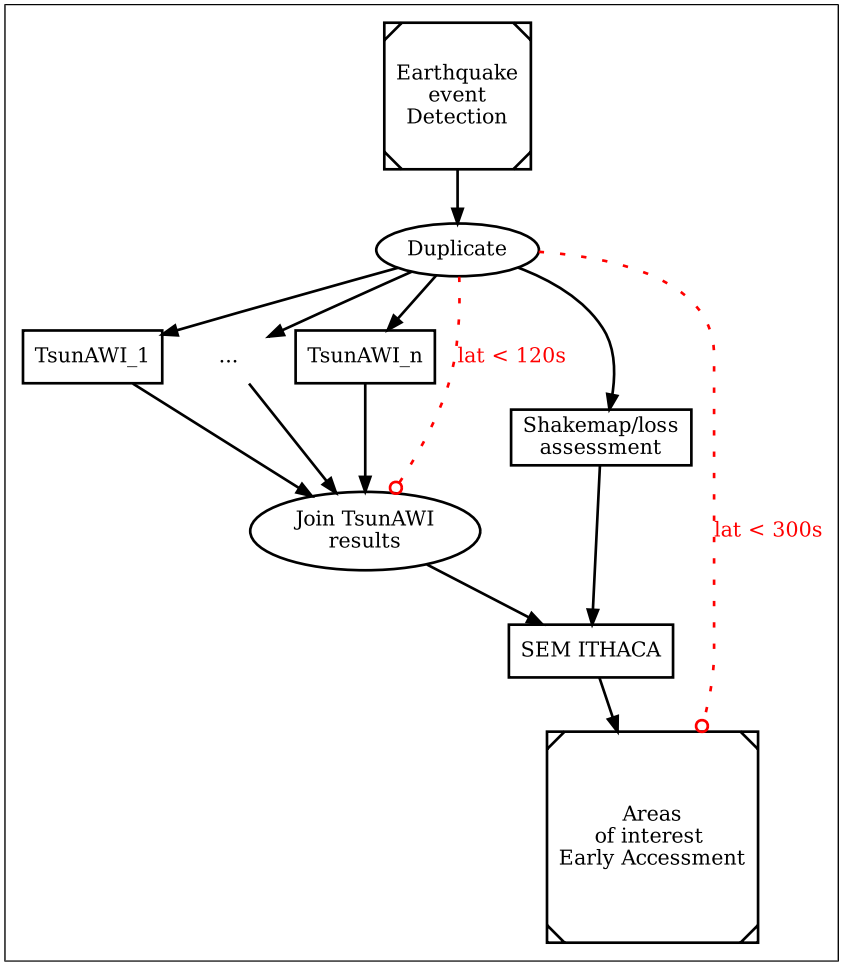

Figure 4. The MoC model for earthquake/tsunami early warning part. In red, the planned latencies as part as the time-constraints. Data-centered processes are ellipse-shaped whereas computing processes are in rectangular (ordinary) or square shape (sources or sinks). The similarities with Figure 3 is quite visible. The high-resolution workflow is similar but is not time-constrained and possesses only one high-resolution TsunAWI process.

event-triggered process and upon detection of a quakeml file (i.e. when a first set of data identifying the magnitude and epicentre of the earthquake is aggregated in a file), the LEXIS platform is put to contribution to run step 2 which consists in two data-paths using the first parameters of the earthquake. The first path (left-hand side of the figure) will run a limited ensemble of coarse-grain TsunAWI simulations. The second path would start the evaluation of a shakemap and damage assessment directly due to the earthquake, in parallel. Both paths can be run on different locations, transparently thanks to the DDI.

The third step is a fusion process that runs after both the end of TsunAWI agglomerated executions and the earthquake damage assessment. This step, developed by ITHACA, is a Satellite-based Emergency Mapping (SEM) aimed at emergency services that, in this particular case, consists in the highlighting of priority areas (based on damage estimations) for an early-tasking of satellite acquisitions to be submitted to satellite data providers. This step aims at improving the delivery time of SEM products and can also act as a first information set to plan the emergency activities (emergency rescues and prevention of further risks).

As shown in the Figure 3 the shakemap evaluation and damage assessment are defined as cloud task, as well as area of interest computation provided by ITHACA. These processes are provided as docker images which are used for the computation. Tsunami inundation computation by TsunAWI is an HPC task which is deployed at used HPC centres and executed using HEAppE middleware. Ystia orchestrator takes care of execution of these tasks at available hardware and also data transfers using LEXIS DDI.

All the particular processes are detailed in their respective sections, later in this paper.

3.3. The data-flow within the workflow

The event-triggered part of the workflow handles earthquake events. It starts upon reception of the initial earthquake event[4], when the hypocenter and magnitude are first known. It initiates a fast compute path that combines fast tsunami simulations with preliminary inundation estimations, and a generation of a shakemap and a fast, aggregate loss assessment. After both paths are completed, which is only time-bounded by the maximum observed execution time of the different processes, the fusion step (step 3) provides the finalized data to relevant services. To that end, our main target scenario, is a M8.3 tsunamigenic earthquake near Coquimbo, Chile. The associated time constraints are fitted under 3 minutes (usually 2 minutes) for the covered area of the coarse-grain tsunami and earthquake simulations.

This simplified workflow including a limited ensemble computation of coarse grain TsunAWI simulations, earthquake damage evaluation, up to the fusion process and impact map generation was implemented and successfully tested on the aforementioned earthquake scenario of Coquimbo/Chile within the specified time-frame, including data movements, container instanciations, and so-on. This validates the principles described in the paper of an urgent-computing realistic utilization.

When talking about a time-constrained workflow, we need to also mention data transfer overhead. There are three critical places in case of the data transfer:

(1) From systems measuring the earthquakes to the computational infrastructure.
(2) From one computational infrastructure to the other in the event that some parts are computed faster on one than the other and there would be a need for aggregation of the results.
(3) Posting the results to early warning systems.

In our experiment, we use HPC centres that are connected by the high-speed Internet in terms of hundreds of Mbit/s during the normal; therefore, even in case of point 2 with the data size in terms of tens of MB we will get quite fast transfers, this is enabled by the LEXIS Platform which uses iRODS[5] to stage data from one centre to another. Thanks to this high-speed internet also, the points 1 and 3 are mitigated to units of seconds, thanks to small size of needed input data and output data, usually tens of MBs, as long as the source of data in case of 1 and receiver in case of 3 are on the reasonably fast line in reasonable distance from the data center. These points can be ensured by using data centers located closer to the points of interest. In case the Cloud partition and HPC partition are located in the same data center the point 2 will become completely negligible since usually these partitions are connected by high-speed connection with a bandwidth of several GB/s.

3.4. Future evolutions

Whilst this first (quick/early and coarse-grain evaluation) data-flow path provides areas of interests in matters of minutes after the first warning of the event, our final aim is to run a second (after the end of the event, slower but fine-grain and more precise) data-flow path that would provide the detailed inputs and detailed areas of interest in matters of a few to several dozen of minutes depending on the size of the monitored area (see Figure 5). On our main Coquimbo, Chile scenario, it would mean around 10 minutes for the detailed quakeml file (with detailed profile, and this must wait for the end of the event with a good probability), then have a second step with fine-grain tsunami and earthquake assessment simulations. That take around 10 to 15 minutes on the LEXIS infrastructure (for the fine grain simulation, which means a typical tsunami grid down to 20 m), with only one fine grained simulation really required.

3.4.1. Running several scenarios and timeliness. The first assessment for TsunAWI's evaluation of inundation levels requires several execution of the TsunAWI simulator with a distribution of parameters to compensate for the low accuracy of the input data: Indeed, knowing only a magnitude and an epicenter can be insufficient to capture a single worst-case scenario. That is why we generate an ensemble parameters to run several scenarios in parallel in this part of the workflow. At the end, the worst inundation levels from the several scenarios are selected.

One question may arise, nonetheless: how can we ensure that some operational results can always be available within the short time-window required by our early warning infrastructure? As the availability of the set of computing resources required by our first stage tsunAWI simulation sets is not guaranteed in a cloud or an HPC center. But as long as at least one result is generated, the workflow provides an output.

So, there are two corollary questions. The first is the availability of a minimum set of computing resources, for which we will provide some hints at the end of this section, and the second being the timeliness of a set of simulations when the number of maximum computing resources is not guaranteed.

3.4.2. Going further on time constraints and reliability. In our workflow, for early warning, typically 1 to 3 minutes is an adequate requirement: allowing early warning of population and reducing both false positives and false negatives.

As shown on Figure 6, the utilization of the Yorc orchestrator can allow to enforce an ensemble simulation (here on a simplified case of the Padang city in Indonesia) with a strict deadline from the instantiation of the container to the output of the simulated results in the DDI virtual filesystem. Theoretically, this would allow: First

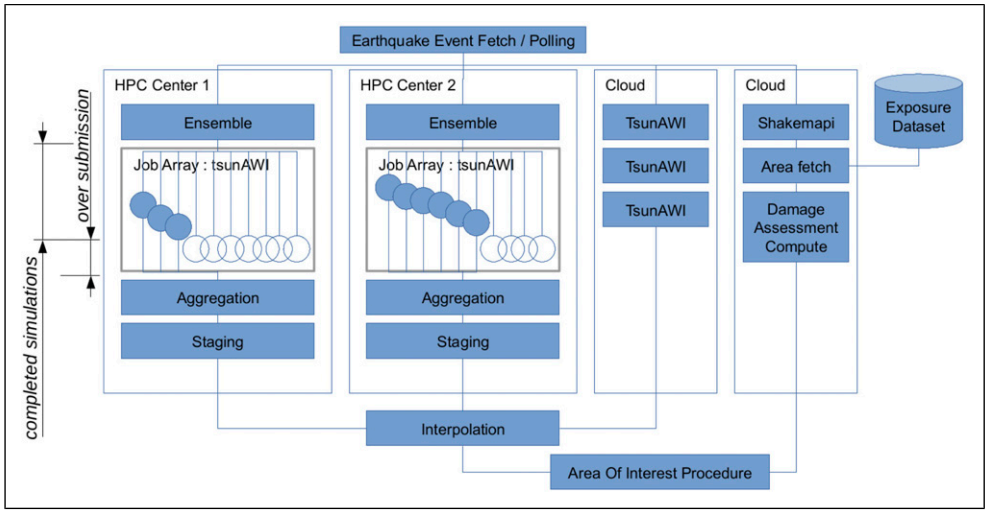

Figure 5. The first stage workflow, from the first earthquake event information to the area of interest determination, showing how two geographically separated HPC centers and cloud resources are combined.

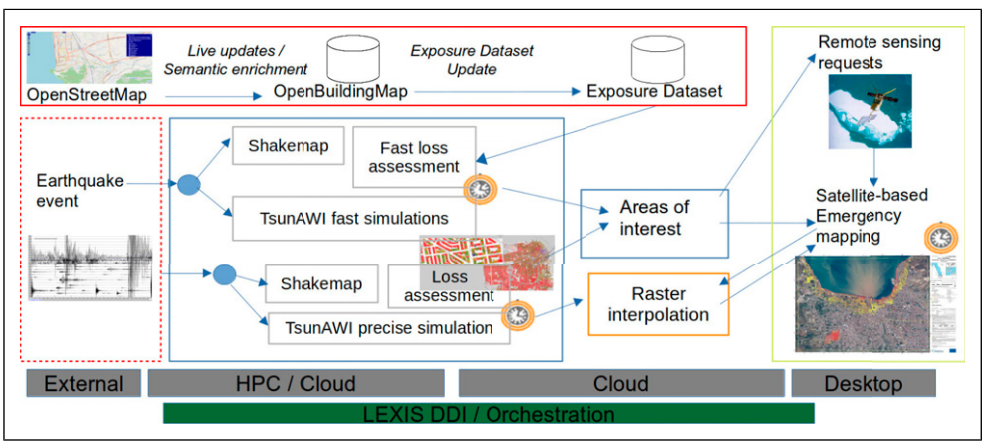

Figure 6. Our aspirational and complete earthquake and tsunami workflow overlaid over the LEXIS orchestration and data infrastructure, with in red the permanently running exposure dataset creation, in dashed line, the event triggered part followed by the compute block (blue), driving, in turn, the satellite-based emergency mapping. It shows two paths for final SEM, a rapid and time-bounded one for early warning and a second more accurate, especially for emergency team management.

an enforcement of deadline on any ensemble simulation (for the Coquimbo test-case, this deadline should be chosen at 2 or 3 minutes); second, the DDI can distribute virtually both the input quakeml file and the simulated tsunami inundation files on several HPC or cloud infrastructures.

The first improves the timeliness of the system, even in the hypothesis of a burdened cloud or HPC center: if not all the simulations can be run before the deadline, there would still be several simulations available at the end of the deadline. The second allows for taking into account when a HPC or cloud infrastructure is either under maintenance or too busy to allow a rapid allocation of computing resources. The combination of both mechanism allows for a high probability of obtaining adequate results within the time constraints. The only strong remaining hypothesis is if the "step 1" process can always run on at least one computing resource at any time. This hypothesis is nonetheless a requirement for all urgent processing applications. Therefore it provides an incentive for specialized services in cloud or HPC infrastructures, but the main point of step 1 is that is a low intensity process that could be part of specialized services of future orchestrators, the same way the deadline enforcement has been provided to the York orchestrator.

3.5. Availability of computing resources

Enforcing time-constraints in such a workflow modeled as a data-flow is required to ensure: first the availability of a minimum set of computing resources, and second that any group of processes will always have a well define deadline and enforce it if required. The first point can require a prioritization of a minimal set of resources ready to be diverted to the execution of a shrunk down version of the workflow that can provide an outcome. Nowadays, there does not seem to have a commercial case for guaranteed on-demand availability of computing resources in cloud infrastructure or HPC centers. Nonetheless, this could change if urgent computing and similar applications become more widespread.

In the meantime, one can argue it is possible to utilize the open-source framework of the LEXIS project to initiate a minimal configuration on a handful of dedicated servers with high availability. Such services are available for Internet web sites, and can be utilized on the basis of the LEXIS DDI and York orchestrator to provide a first minimum batch of results. This would guarantee the enforcement of time constraints, since the simultaneous unplanned unavailability of more than 2 high availability servers would have an extremely low probability[6]. Additional granted access to cloud infrastructures can be used to provide more computing resource to the urgent computing workflow. While not mandatory in this scenario, this would enhance the results and lower the uncertainties on the early warning areas.

In that utilization of the LEXIS software environment, we can think of this kind of application as a cloud-assisted urgent-computing workflow. As mentioned previously, such use cases can constitute a potential future market for cloud providers.

In our case, we tested the workflow on the LEXIS Platform deployed at IT4Innovations, Czech National Supercomputing Centre[7]and Leibniz-Rechenzentrum der Bayerischen Akademie der Wissenschaften[8] and used their HPC and Cloud instances. The disadvantage of the HPC setup is the use of the scheduler, which does not guarantee instant availability of the resources in case of an event. In fact, for the jobs to be scheduled, it would most likely take quite a long time. To alleviate this problem, it is possible to have multiple centers, combining HPC and Cloud executions, setup to run the workflow, thanks to the LEXIS Platform, and proceed with the quickest results; however, this will still not ensure that the computation itself will be executed in given time constraints.

The best solution to ensure that the physical resources for the computation are available when needed is to change the policies of the HPC centres to allow stopping the running computations in case of the disaster events and execute the workflow instantly with highest priority. This solution is currently impossible, as it goes against the policies of the HPC centres and would most likely also require a change on the legal level to allow HPC centres to stop computations for life-saving computation in case of disaster. Unfortunately, as scientists, we are not able to enforce these changes ourselves. Until then, our solution with overallocation of resources and continuation with the quickest branch is the best possible.

4. Tsunami simulations with TsunAWI

In this section we discuss the implementation of TsunAWI, the tsunami simulation. The general focus is, in the context of urgent computing and its requirements, it is necessary to go two folds: First one must make sure to have a good and as fast as possible tsunami simulation framework. Second, once one is close to the best performance possible both in accuracy and execution time, it would be required to select the granularity of the simulation to retain as good an accuracy as possible while execution is a fraction of the time-frame allowed in the real-time execution framework. These are the so-called *TsunAWI fast simulations* as showcased on Figure 6.

Indeed, if the execution time is on the same time scale as the time-constraints of the urgent computing then there is a high probability that the deadline will be missed. If the execution time is typically a fraction of the time constraint, then it is possible to run a ensemble of simulations allowing to process with a collection of hypothesis. Such a mechanism, as was explained previously, can compensate the fact that the urgent computing will be run with only limited knowledge of the real parameters of the earthquake except for the epicenter and the magnitude. At the end of the ensemble computing, it is possible to proceed to several statistical analyses to trigger alerts or warning to populations.

4.1. Tsunami simulations

The tsunami simulations in the workflow are performed with TsunAWI. The development of this finite element tsunami model started in 2005 as part of the GITEWS project (German Indonesian Tsunami Early Warning System). The model is derived from the global ocean model FESOM Danilov et al. (2004) and inherited the spatial discretisation with unstructured triangular meshes, allowing for ample flexibility concerning realistic coastlines and bathymetric features.

In the first moments, after an earthquake has generated a tsunami, there is little information on the source of the event. Moreover, data on wave propagation is obtained only by a small number of sensors like tide gauges or buoys after the wave has already reached their locations. Therefore, numerical tsunami models can contribute considerably to a complete picture of the potential wave propagation. They can also estimate warning products like arrival times in specified locations and estimated wave height at the coast.

The quality of such estimates strongly depends on the available data on the tsunami source. In the case of earthquakes, initially, only estimates of magnitude and epicentre location are available. After several minutes, more details like hypocenter depth or moment tensor become available, refining the initial conditions for the tsunami simulations. HPC systems reduce the time needed for single simulations and allow for parallel calculation of tsunami ensembles covering a range of source parameters and accounting for the related uncertainties.

The situation is exemplified in Figure 7. The left panel shows the initial elevation derived from an inversion of the the event in September 2015 based on a large number of seismic recordings published in Shrivastava et al. (2016). Such elaborated source models are valuable for verifying tsunami simulations. However, the necessary processing takes time, and first estimates need to rely on much more simplified sources. One example is shown in the right panel of Figure 7. The depicted initial sea surface elevation at the epicentre location is symmetric and cosine shaped. The size and amplitude are calculated from moment magnitude Mw = 8.3 from the scaling relations by Strasser et al. (2010).

Figure 8 shows the triangular mesh in Coquimbo Bay, where triangles cover the land part in the priority area with an edge length of about 150m. We employ this mesh for the fast simulation, while more detailed simulations on a mesh with 20m resolution follow as soon as the source of the tsunami is better determined. The resolution in the deep ocean is about 10 km, whereas the resolution in the coastal area is considerably finer to account for the run-up process.

The inundation displayed in Figure 8 was calculated in the coarse mesh for the idealised source (Figure 7 right). The corresponding values in the triangular mesh are interpolated to a regular raster for a simplified exchange of data products with project partners in standard formats like GeoTIFF files. These products are used for estimates of the tsunami impact, as described in the following sections.

4.2. TsunAWI code optimisation and parallelisation

Originally, TsunAWI was designed to pre-compute scenarios and provide a database of warning products and inundation estimates. Until recently, real-time simulations were limited to the simplified model physics (e.g. linear shallow water equations, no inundation) discretised on coarse, regular grids, which is sufficient to assess the wave height and the arrival time at the coast. First examples for highly optimised codes also addressing the inundation in real time are HySea Macías et al. (2017) and Volna-OP2 Giles et al. (2021), while the MOST framework relies on a multi-step approach, as described in Tang et al. (2016) for the Chilean tsunami on 16 September 2015.

Throughput and ease of use of TsunAWI for external users were more important than time to solution. Consequently, we adhered to an OpenMP parallelisation and spending some effort on optimising data structures and the most dominant loops.

- Loops with dependencies that inhibit OpenMP parallelisation were restructured and parallelised. E.g., instead of looping over the edges and adding a contribution to each adjacent vertex, the loop is now over the vertices collecting the contribution from all adjacent edges.
- To avoid multiple computations of constant values, auxiliary fields with geometry-dependent values are precomputed.
- Linear model physics on multi-core architectures results in a runtime that is mainly limited by memory bandwidth. Therefore, cache reuse was a significant concern, e.g. by fusing loops and carefully choosing the memory layout of fields, e.g. It was beneficial to join the arrays containing the West-East and the South-North components of the velocities because their accesses are (always) joined.
- OpenMP on NUMA (non-uniform memory access) architectures makes data locality even more crucial. First touch and static scheduling account for this.
- Both cache and NUMA benefit from well-sorted mesh vertices and elements. Sorting along a space-filling curve reduces cache misses, access to remote memory, and false sharing.
- Last but not least, most tsunami simulations impose a high load unbalance, as, in regions not yet reached by the propagating tsunami, all values remain zero and can be masked from the computation. In particular, the resolution on land is high, so many mesh vertices are never reached. The OpenMP dynamic scheduling seems an obvious answer. However, it does not pay off, as it destroys data locality.

With the above improvements and hardware development, real-time simulations came into reach for TsunAWI. LEXIS allowed us to focus on the topic. The first approach was to determine how coarse a mesh can be to still allow for a proper estimate of the inundation in a regional focus. Previous studies Griffin et al. (2015) showed that the region of interest should be discretised with a resolution of 20 m to obtain inundation maps that are suited, e.g. for evacuation planning. For a first rough assessment, however, 150 m-200 m is sufficient.

Then, we picked one of the last remaining low-hanging fruits and addressed the limiting memory bandwidth by switching the arithmetic from double to single precision. This modification speeds up the simulations, however the effect decreases with increasing parallelisation, as the subdomains become so small that all values fit into the cache. Only the reduced MPI message size remains an

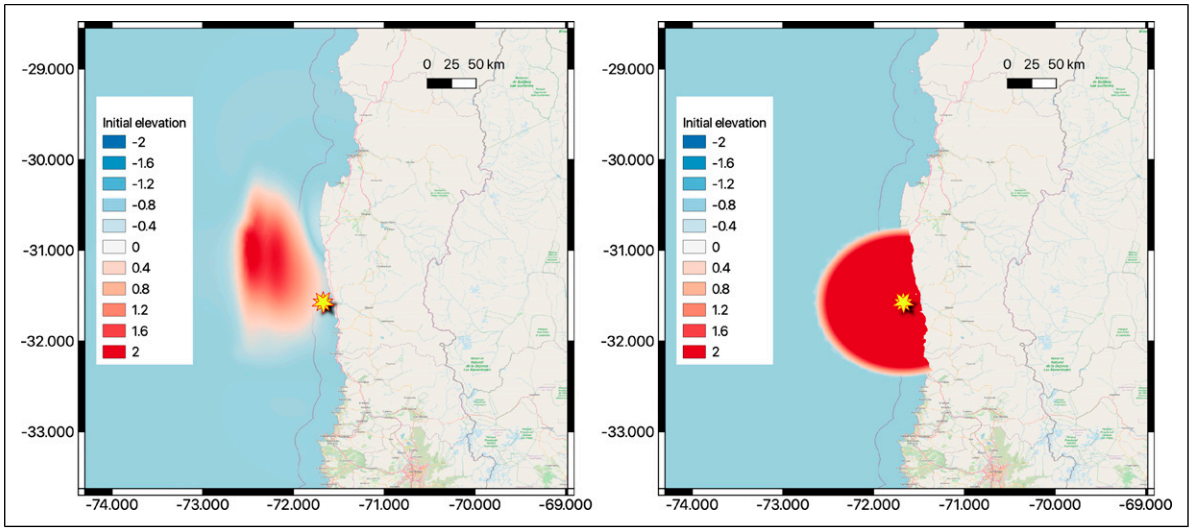

Figure 7. Source region of the tsunami event offshore Chile on 16 September 2015. The epicentre location is marked by a star. The left panel shows the initial surface elevation derived from the detailed source model by Shrivastava et al. (2016), whereas the right panel depicts the simplified source based solely on the epicentre location and a symmetric cosine-shaped initial elevation derived from the moment magnitude and the scaling law by Strasser et al. (2010). Background pictures: © OpenStreetMap contributors.

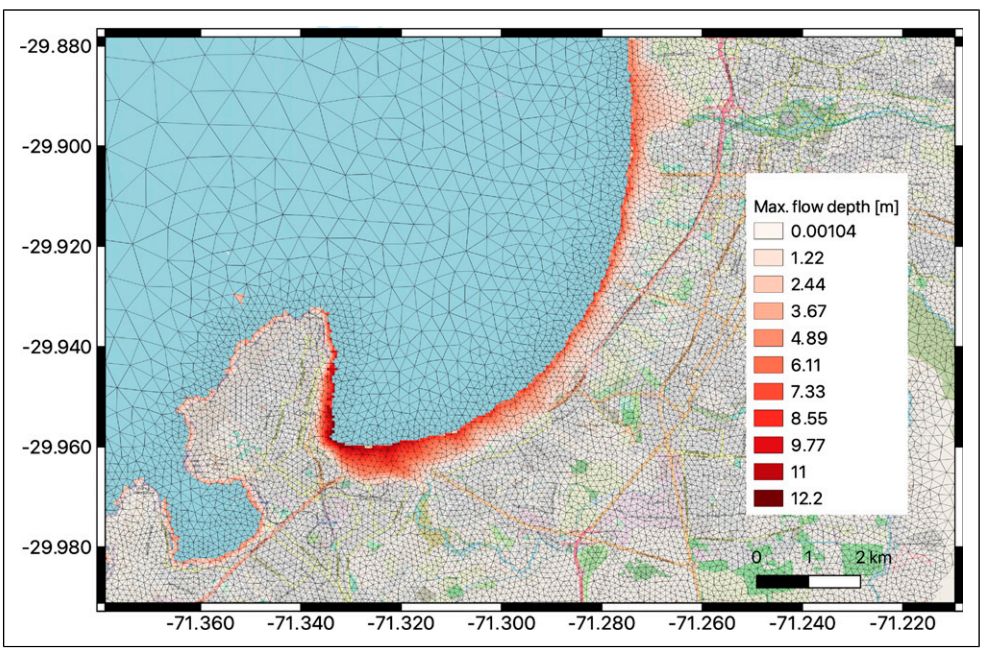

Figure 8. A small section of a triangular mesh with approximately 150m mean resolution in Coquimbo and La Serena areas, the coarse regional setup for the Chilean 2015 event. Also shown is the inundation area obtained from the simulation with the idealised source depicted in the right panel of Figure 7. Background picture: (c) OpenStreetMap contributors.

advantage of the single precision. This was a good choice for shallow water equations and improved the simulation time on a single compute node by about 30%, see Table 1.

The most important achievement for TsunAWI in the LEXIS project was to add MPI parallelisation. We adapt the framework used in the global ocean model FESOM, which shows excellent strong and weak scalability Koldunov et al. (2019).

The basis is a domain decomposition with the graph partitioning software metis Karypis and Kumar (1998). The

decomposition, which only depends on the computational mesh, is performed in a preprocessing step and is reused for all simulations with the same number of MPI tasks.

Within TsunAWI, three types of MPI communication are employed.

- Exchange of values on "halo" vertices and elements. Each MPI task holds parts of the mesh and the adjacent halo vertices and elements. Values on the interface to the neighbouring subdomains must be exchanged during each time step. Values on the edges are derived from the other halo values and do not need to be communicated.

As a start, we employ non-blocking MPI_Isend, MPI_Irecv to overlap the communications with all adjacent subdomains and, if possible, also with independent computations.

- In TsunAWI, global collectives are only required for diagnostic purposes, such that non-blocking calls like MPI_I(all)reduce can overlap with a whole time step and do not form a bottleneck.
- Last but not least, global fields must be read and written. This is implemented in a serialised way and can dominate the time to solution. The master task reads most of the input data and scatters it to the other processes. Only the mesh decomposition is stored in one file per task and is read in parallel. Finally, the master collects and writes the output. For further optimization, the I/O should be performed asynchronously, overlapping with computation.

The run times for typical TsunAWI scenarios are listed in the Tables 1 and 2. The regional coarse and fine resolution scenarios for the Coquimbo and La Serena areas, Chile, and Padang in Sumatra, Indonesia, were set up for the LEXIS workflow. However, with the MPI parallelisation, we can tackle basin-wide scenarios in real-time. The setup "Indonesia" was used to compute 10,000 scenarios for the tsunami scenario database at the Indonesia Tsunami Early Warning Centre. The computational mesh spans the Indonesian Archipelago from 88° East to 142° East and 22° South to 19° North and resolves all coastal regions with 250 m and areas of larger cities with 150 m resolution.

Keeping the OpenMP directives, TsunAWI can also run in hybrid mode with both MPI as the first level of parallelisation and OpenMP on the task level. The first tests gave mixed impressions. Generally, a low number of OpenMP threads, 2-6, was beneficial in employing hyperthreading. However, the expectation of mapping each NUMA domain to one MPI task and tackling the load unbalance with OpenMP dynamic scheduling could not generally be met. Also, we did not always observe better scaling by reducing the number of MPI tasks for the same number of compute nodes. Furthermore, hybrid code needs careful handling to map each task and thread to reflect the cluster architecture. Given enough time to prepare a large set of scenarios with one setup, tuning hybrid runs is worth the effort.

Summing up, with the MPI parallelisation and sufficient compute resources at hand, real-time simulation of tsunami propagation and inundation with TsunAWI is no longer restricted to small areas and/or coarse resolution. Porting TsunAWIs to FPGAs was also experimented in the project Lenormand et al. (2022). Such feature would give even more flexibility to integrate TsunAWI into workflows with constraints on throughput and time to solve.

The MPI parallelisation is a work in progress, with an initial careful approach, adding both OpenMP and MPI synchronisations generously. More work has to be done. Less used features of TsunAWI have to be added to MPI, e.g., writing raster output and tide gauge data on the fly. Most importantly, the I/O is still serialised and should become asynchronous. The load

Table 1. TsunAWI setups for the chilean event 2015 and they're respective compute time for the time stepping, measured on the HLRN (North German supercomputing alliance) system "Emmy". Each computed node consists of §2× intel cascade lake platinum 9242 (CLX-AP) with 48 cores each.

Setup		Coquimbo region coarse	Coquimbo region fine
Number of vertices		1,709,500	4,888,000
Resolution		150 m–15 km	20 m–15 km
Time step dt		2.0 s	0.15 s
Model time		10 h	10 h
Compute time for time stepping, single/double precision			
1 node	96 tasks	93s/133 s	3070s/4350 s
2 nodes	192 tasks	50s/72 s	1760s/2465 s
4 nodes	384 tasks	27s/32 s	945s/1330 s
10 nodes	960 tasks	13s/14 s	378s/452 s
20 nodes	1920 tasks	9 s/9 s	224s/254 s
40 nodes	3840 tasks	–	149s/156 s

Table 2. Three different TsunAWI setups for Indonesia and their respective compute time for the time stepping, measured on the HLRN "Emmy".

Setup	Padang region coarse		Padang region fine	Indonesia
Number of vertices	230,000		1,240,000	11,110,000
Resolution	200 m–15 km		20m–5 km	150 m–20 km
Time step dt	1.5 s		0.15 s	1.0 s
Model time	2 h		2 h	12 h
Compute time for time stepping, single precision				
1 node	96 tasks	4.0 s	182 s	3045 s
2 nodes	192 tasks	2.5 s	99 s	1595 s
4 nodes	384 tasks	–	55 s	820 s
10 nodes	960 tasks	–	27 s	317 s
20 nodes	1920 tasks	–	17 s	166 s
40 nodes	3840 tasks	–	–	96 s

unbalance can be tackled by a weighted partitioning, e.g., by the distance to the epicentre or height above zero. The MPI communication must also be tuned, e.g., by better overlapping communication and computation, which is prepared but not yet fully employed; by persistent MPI for repetitive communication in the time stepping scheme; and by using the knowledge on the topology of the multi-core nodes and interconnects with neighbourhood collectives. The MPI-4 standard defines persistent neighbourhood collectives like MPI_Alltoallv_init, but it has yet to be added to MPI implementations.

5. Earthquake loss assessment

As in the previous section regarding TsunAWI, the tsunami simulation, in this section we describe the general approach to earthquake simulation as implemented by GFZ but in the goal of showcasing the part of the *Fast loss assessment* as presented on Figures 6 and 4 to be compatible with the real-time constraints. For this early warning to populations, we want to focus more on timeliness than on accuracy, as it is necessary to obtain the results before the deadline.

The state of the art in earthquake simulations as can be found in Paolucci et al. (2014), McCallen et al. (2021), are important to our understanding on how a given earthquake impacts infrastructures, but such a process requires both accurate and detailed data on the earthquake profile and the buildings, and a long simulation time with huge amount of processing power. Both conditions are incompatible with this urgent computing workflow. This is why in this section we focus on a earthquake simulation framework adequate with regards to the imposed time constraints.

5.1. General earthquake simulations

Any Earthquake and tsunami loss assessment requires a combination of an impact quantification of the event (earthquake and/or tsunami), an exposure model and corresponding fragility functions. The impact quantification for the tsunami, the tsunami modeling, is described in the previous section. For earthquakes, impact is usually given as a ground-motion field, describing the spatial distribution of ground motions. These ground motions are then combined with the exposure model that describes the different types of buildings, their various properties, value, and the number of people expected inside by computing the probabilities for each building to encounter each of the damage states (from no damage to complete destruction) based on its building type.

Within LEXIS, we have optimized the ground-motion field generation and provided it as a fast web service. The exposure model for the loss assessment is based on building data from OpenStreetMap. We have undertaken multiple steps to reduce the computational times for the exposure generation and exposure-data delivery.

5.2. Ground-motion field generation

We developed a web engine to generate synthetic ground-motion fields based on codes of the OpenQuake engine of the Global Earthquake Model (GEM) foundation (https://www.globalquakemodel.org/openquake). The backend asynchronously digests requests via an API using earthquake source parameters (depth, epicentral location, moment magnitude and focal mechanism). It returns a ground-motion field in user-defined intensity measures (e. g. peak ground acceleration or intensity) and various formats such as ASCII, GeoJSON, etc. An interactive interface allows one to explore the expected spatial ground-motion distribution by selecting locations of interest on a map and defining the earthquake source interactively in a web browser.

This tool is used to generate the earthquake impact measure for the loss assessment; it is freely available at

https://git.gfz-potsdam.de/loki/shakemapi. A repository has been created to host representative scenarios which have been designed and made available to all project partners at https://git.gfz-potsdam.de/loki/lexis-scenario-examples.

5.3. Processing building data

The exposure model to be used in rapid loss assessment is based on building data from OpenStreetMap (OSM).

5.3.1. Quadtree. Requesting data on the building scale from a global database is computationally costly. To ensure fast performance in delivering building data for rapid loss assessments and to keep the database as small as possible, the data is prepared on a geographic tile basis, i. e. all information of buildings with their centroid in the particular tile are stored in that tile.

To implement a computationally efficient and easy-to-use tile grid, We are adopting the Quadtree data structure approach Finkel and Bentley (1974) to define a tile grid across the planet in Web Mercator projection (EPSG:3857) with a target tile size at the zoom level 18. Using an algorithm to aggregate tiles (cascading) into their parent tile allows us to reduce the total number of tiles by a huge margin. Water tiles or empty tiles with no relevant data can be aggregated upwards the hierarchy of the Quadtree as long as all child tiles of the parent tile are empty or contain water.

Using this framework of aggregation and de-aggregation, we calculated a coastline multi-resolution tile grid from OSM coastline vector data. Starting at zoom level 0, we de-aggregated every tile up to zoom level 18 if it contained a coastline, resulting in a set of largest possible tile sizes for tiles covering water or empty land only, while the coastline is described in high-resolution with zoom-level 18 tiles. This de-aggregation reduces the total number of tiles in our database from approx. 68 billion tiles for a zoom-level 18 only grid to approx. 100 million tiles, see the two globe insets in Figure 9.

5.3.2. Tile completeness estimates. Although OSM contains close to half a billion footprints as of now, building coverage is not complete everywhere. As a consequence, we need to have completeness assessments for each tile. A tile is considered complete when all real-world buildings are mapped in OSM, otherwise incomplete. Additional classes cover unknown for tiles that have not yet been assessed, undecidable when no ground truth can be determined (e. g. due to cloud coverage in the satellite imagery), water or empty when no buildings are present.

To create these completeness data, we are following a multi-tier approach:

- We built a website on which an analyst can set the completeness status for one or more cells by comparing the satellite imagery with the overlay of

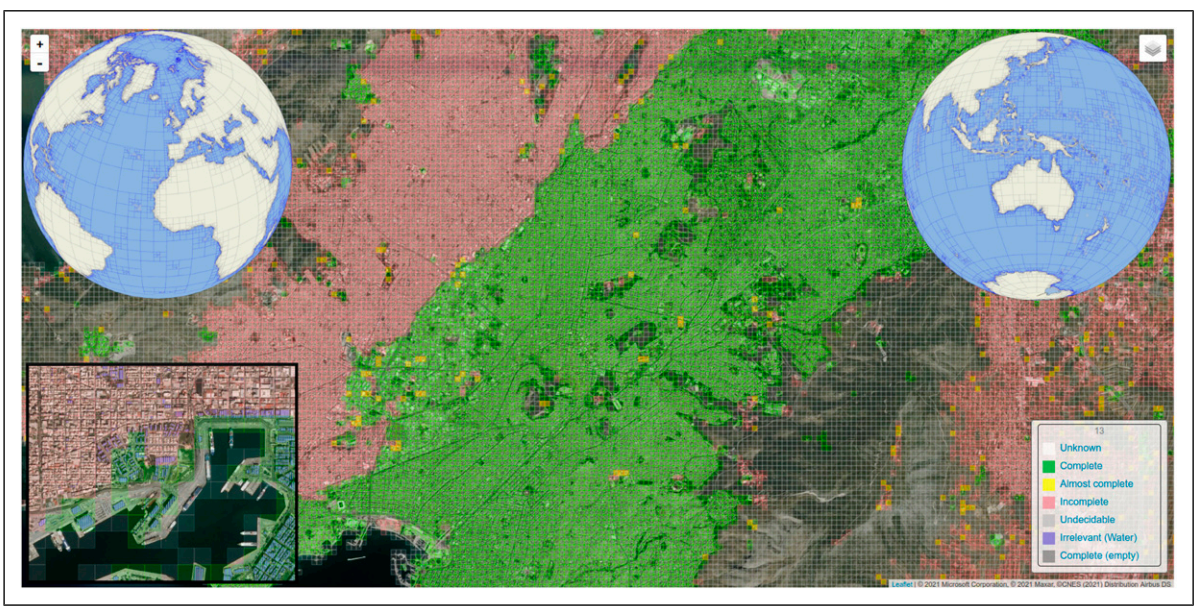

Figure 9. Quadtree usage for the building data. (Background figure and bottom left inset) Completeness data at zoom level 18 in the area around Athens, Greece, displayed by the Clickpleteness. The Clickpleteness is a web application that enables users to set the completeness status (color coded) of tiles by comparing an OSM building layer against underlying satellite imagery, assessing whether all real-world buildings (blue shapes in the inset) are mapped in OSM and assigning a degree of completeness (see legend in the bottom right inset) to that tile. (Top left and right globes) Quadtree grid resolving the coastline data globally. Tiles covering only sea water or land (including inland waters) are aggregated to the maximum size, i. e. lowest possible zoom level. Data and map copyright by OpenStreetMap and contributors.

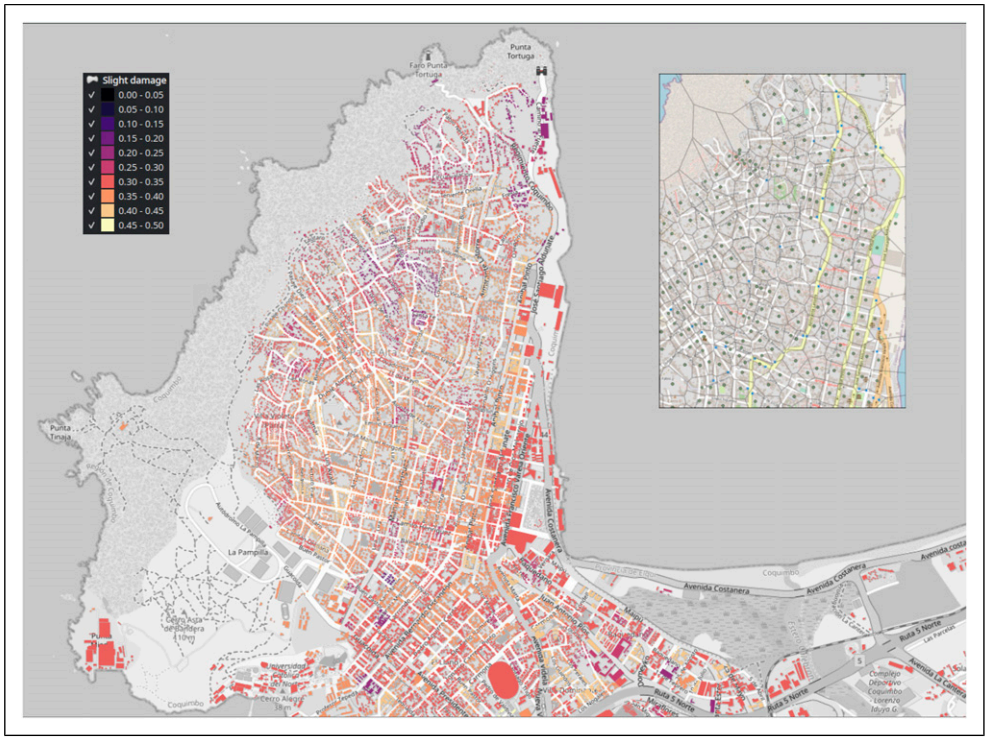

Figure 10. Distribution of probabilities (color coded) for buildings to experience slight damage due to shaking as caused by the 2015 M8.3 Coquimbo earthquake. (Inset) Map of the northern part of Coquimbo, with the data points of the SARA model (green circles) and their respective Voronoi cells (gray polygons). Data and map copyright by OpenStreetMap and contributors.

building polygons directly, see bottom-left inset in Figure 9.

- In a collaboration with the team at the University Heidelberg/HeiGiT, we included a completeness analysis into the MapSwipe smartphone application Ullah et al. (2021). This application crowd-source the comparison of building footprints with satellite imagery, generating a completeness estimate per tile.
- The last approach is less accurate but can be employed automatically and globally. With remote sensing techniques based on mainly Sentinel-I radar data, we can identify built areas. Preliminary investigations in areas with complete building stock data show useful correlations between the size of the built area and the size of the building footprints such that in absence of more detailed information, the remote-sensing estimate of built area can be used as proxy to number of buildings per cell.

5.3.3. Example damage assessment. We developed a residential exposure model for the city of Coquimbo, Chile, based on the aggregated exposure data from the SARA project Yepes-Estrada et al. (2017). This model describes aggregates of building types per location. To distribute these aggregates to the buildings from OSM, we defined Voronoi cells (see inset in Figure 10 and associated every real building to its nearest SARA data location in order to compute the building type distribution from the data given by the SARA model. The preliminary model treats each building as residential for simplicity but can be expanded to cover the other occupancy cases.

We developed a damage calculator to work with our exposure model. The calculator interpolates the ground-motion value for all asset locations (either the center of each tile or a single building). From the taxonomy of each asset, the calculator retrieves the corresponding fragility function and computes the probabilities for each possible damage state. These damage states can be visualized directly at the building resolution (see Figure 10 because all building geometries are processed together with the asset data.

5.3.4. New processing engine. Generating a model that contains almost half a billion buildings and their related exposure indicators and is based on different data sources such as the specially pre-processed OSM data and satellite imagery requires a scalable and modular processing engine. To tackle the challenges of Big Data computing with continuously changing heterogeneous data sources, we designed Rabotnik. It is a Python-based framework for distributed, real-time processing of large datasets, which makes it easy for scientists to define their own processing rules.

The engine is built around atomic computational units which we call rules that form the entry point for users of the Rabotnik framework. Each rule contains the logic and analysis a programmer wants to execute when one datum is changed. For example the completeness of a tile has been assessed and updated. The update is detected and will trigger a signal that is being broadcasted into the Rabotnik system. Rabotnik builds on top of existing, solid and state-of-the-art open-source solutions like Celery and RabbitMQ, and uses the Message Queuing Telemetry Transport (MQTT) protocol to broadcast messages and to send data packages between client instances. The emitted signal is sent to a message broker (RabbitMQ) and from there dispatched to clients that are registered to receive the designated signal.

Each updated, added or removed datum needs to be reflected within the different analyses which all together constitute the exposure model. The highly complex dependencies between individually developed and maintained data-analysis stages requires a flexible and extensible task queueing, processing and messaging system that is scalable across computational nodes to enable a real-time processing of the ever changing datasets.

The decentralized architecture allows any participant to subscribe to broadcasting channels and to trigger computations. The receiving instance can in turn issue a message for other subscribed entities to react upon. When receiving a signal, a client instance executes the contained rules. Despite versioning the rules, this level of abstraction also allows to track the rules that contributed to generating one datum of the exposure model. This, in turn, also enables us to identify rules that need to be re-executed when an input datum is changed. This allows us to efficiently identify data that needs to be updated without the need of full re-calculation.

Rabotnik harnesses the Celery framework which allows distributed execution and a high degree of scalability. To avoid blocking code execution in this CPU- and IO-intensive processing, Rabotnik operates fully asynchronously. To further enhance coordination of tasks the message broker organizes the dispatching of messages to clients in message queues. As some tasks are more important than others the queue management allows to set different priorities of message queues. If for example a catastrophic event happens, this allows to focus computation on the area of interest to support a near real-time re-evaluation of the situation.

The current work in progress is published as Free Software, under AGPLv3 + license. The source code is available at: https://git.gfz-potsdam.de/dynamicexposure/rabotnik/rabotnik.

6. Satellite-based emergency mapping

Satellite-based Emergency Mapping (SEM) mechanism provides information derived by the analysis of satellite imagery for emergency response in relation to different types of natural disasters, including earthquake and tsunami.

The Copernicus Emergency Management Service (CEMS), the European Commission SEM service, covers all the phases of the emergency management cycle, from the satellite tasking and image acquisition until the delivery of maps required by the user. With its 24/7 availability, it provides relevant information a few hours following the event. Since timeliness in providing data is the major constraint in the response phase immediately after a disaster, SEM service can take great advantages using automatic models, procedures and HPC processing.

The integration of the outputs provided from simulations of earthquake and tsunami events, with proper exposure information, can effectively support CEMS Rapid Mapping (CEMS-RM) activities, helping to reduce both cost and time for operations. In particular, the following applications have been envisaged:

(a) a supporting function to the early identification and prioritization of areas where to trigger satellite acquisitions and perform satellite-based damage assessment. Satellite images are the main source of information and early tasking of satellite acquisitions can improve the delivery time of the SEM products. A fast identification of potentially affected areas obtained integrating the earthquake loss assessment dataset and tsunami model outputs with exposed assets, as well as the identification, where possible, of the expected damage level, has been proposed in this frame to guide the satellite image acquisition process. The potential benefit of a satellite early tasking based on early warning information compared to the standard workflow for satellite acquisition has been highlighted by Ajmar et al. (2019). For this purpose, particular attention has been paid to the definition of an automatic procedure supporting the identification of potential areas of interest to be submitted to the satellite data providers for post-event imagery supply operations, avoiding or reducing any delay.

(b) the automatic generation of a specific CEMS-RM First Estimate Product (FEP). A FEP is an early information product which aims at providing an extremely fast (yet rough) assessment of most affected locations within the area of interest https://emergency.copernicus.eu/mapping/ems/rapid-mapping-portfolio (https://mapping.emergency.copernicus.eu/about/rapid-mapping-portfolio/). Different are the purposes for FEP production, for instance, they can be used to highlight possibly affected areas, and to set and review the main specifications (product type, areas of interest) of CEMS-RM products to be further

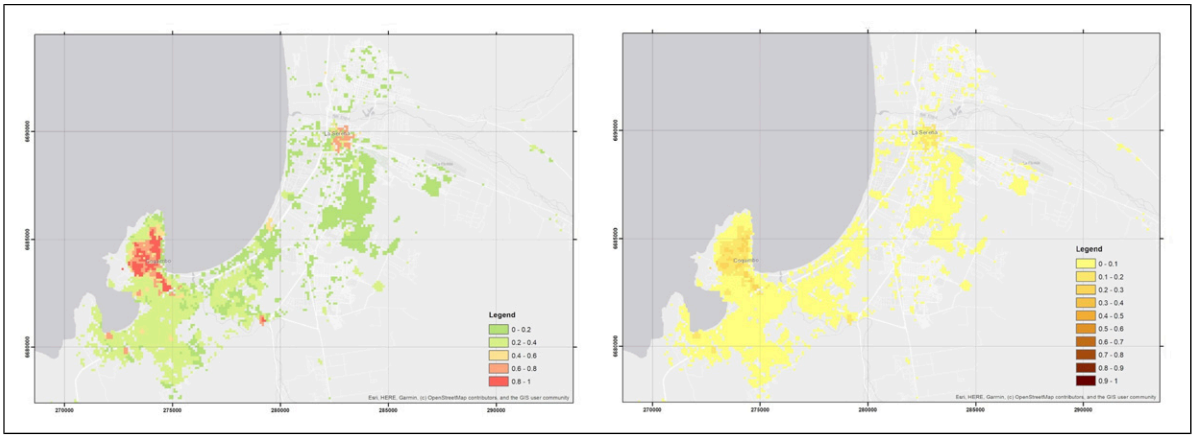

Figure 11. Prioritization of areas of interest. On the left priorization based on the relative potential damage map. Coquimbo and La Serena areas, two of the main Areas of Interest (AOIs) requested by the users during the CEMS activation (EMSR137), are well identified. On the right prioritization based on damage absolute probability map. In this map, for Coquimbo and La Serena areas a reduced level of criticality is visible, which has been confirmed by satellite-based damage assessment operations carried out in these areas in the frame of the CEMS EMSR137 activation and by information concerning damage to buildings, reported in various field survey results (around 60 houses were destroyed and damaged nearly 200, mostly in the Region of Coquimbo; see, for instance, https://reliefweb.int/node/1177936/).

produced. In this context, the assessment and mapping of the most affected assets and FEP production based on information about hazard and damage provided by simulation codes, could allow to obtain different advantages from an operative point of view. Indeed, the procedure are faster than those exploiting satellite imagery, and furthermore highly reproducible and objective.

The following lines describe the cartographic products proposed, in order to support the automatic definition of the most affected areas and FEP generation. The automatic procedures generating them have been developed taking, as a representative use case, the Chilean earthquake and tsunami events occurred on the 16th of September 2015, that triggered the activation of CEMS-RM (https://emergency.copernicus.eu/mapping/list-of-components/EMSR137). In detail for earthquake event, exposure data about population and buildings and probabilities of expected damage information, calculated at a single building level by means of specific models, have been integrated and subsequently aggregated at a proper map scale, suitable for regional level analyses. This aggregation allows the generation of cartographic products for the easy prioritization of affected areas for the eventual satellite tasking operations. For this purpose, three different map products are proposed:

- Relative Potential Damage map, providing the indication of expected most affected areas;
- Damage Absolute Probability map, providing the spatial distribution in the examined area of the absolute values of probability of damage derived considering the most severe damage classes;
- Night Population map, providing an estimation of population exposed to the event in the examined area, based on the population occupancy during the night of buildings.

The Relative Potential Damage product (example in Figure 11) gives an indication of areas where a higher level of damage is expected, integrating both building damage probabilities and the exposure information about building occupancy. The main aim of this map is to help users to identify the spatial distribution of the expected most affected areas. The major limitation of this kind of map is related to its inability to provide an absolute estimation of the expected damage impact level, also usually required by the decision makers for completely depicting the situation and for the immediate prioritization of intervention operations.

All maps are produced based on automatic procedures generating raster outputs using a grid-based approach. In order to resample the information at a building level to the grid level, a conservative approach is adopted. In particular, in order to assign a value to a grid raster cell, the maximum value among all buildings included in each raster cell is assigned to the cell. This procedure provides the largest areas that may be subject to damage, therefore, in the Relative Potential Damage and Damage Absolute Probability maps, the risk of not subjecting potentially affected areas to further SEM is minimized.

In order to correctly support users in the prioritization of areas for tasking satellite acquisitions, the Damage Absolute

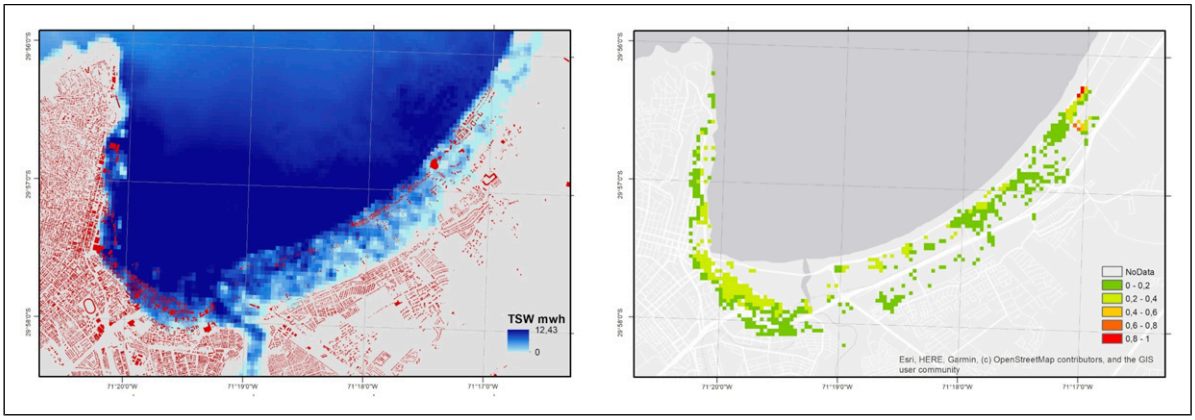

Figure 12. Prioritization of areas of interest based on the potential damage map. Detail on the Coquimbo area. Estimated highly potentially affected areas in the North-West part of the image are due to the presence of edifices (in red in the left image) characterized by high values of occupancy.

Probability product (Figure 11) is also proposed, to be used in addition to the Relative Potential Damage map and covering different user needs. The Damage Absolute Probability map provides, for each raster cell, information about the probability to have severe building damages. This map is estimated taking into account the maximum values of the sum of the probabilities of having "Extensive" and "Complete" damages, supplied in the earthquakes loss assessment dataset derived from simulations. The main aim of this product is helping users to identify areas which are characterized by higher levels of expected damage to be used for the immediate prioritization of intervention operations. Differently from what proposed for the previous product, in this approach population exposure information and the probabilities to have less severe damage levels are not taken into consideration.

The information about building-specific damage assessments caused by earthquakes provided by simulations integrated, at a building level, with exposed population data, and exploited for the generation of the Relative Potential Damage product, have been used to derive a proper CEMS-RM FEP, available in a vector format.

As far as tsunami event is concerned, exposure data about population and buildings, and the maximum wave height expected over inundated areas provided by simulations with TSUNAWI, have been integrated and subsequently aggregated at a proper map scale suitable for regional level analyses, supporting the prioritization of areas for tasking operations and the creation of a FEP.

As a starting point for all products proposed for the tsunami case, the maximum wave height values have been provided at building scale for all edifices belonging to inundated areas, together with exposure data about population occupancy. These data are provided by an automatic procedure calculating summary statistics of these values for each vector geometry overlapping the raster inundation map. Obtained vector building features affected by the inundation, including information about people potentially affected and the expected wave height to which they could be subjected, constitute the FEP based on simulated data.

Furthermore, the described information collected at a building level has been integrated and aggregated at a regional scale generating different raster outputs. The obtained maps give an indication of areas potentially affected by the event and helps also to identify those areas where a higher level of damage could be expected, integrating exposure information about possibly affected buildings, their occupancy and the expected maximum wave height. In particular, two different map products are proposed:

- Potential Damage map (Figure 12), providing the indication of expected most affected areas. This product gives an indication of areas where a higher level of damage is expected, integrating both the wave height values over inundated areas provided by simulations with TSUNAWI and the exposure information about the occupancy of affected buildings;
- Building Exposure map, providing an estimation of the number of buildings exposed to the event in the examined area.

7. Conclusion and outlook

This paper provides a detailed description of the prototype of an Urgent Computing case for Emergency Management in the advent of an earthquake, especially in areas where it can also be tsunamigenic. We have shown that it is possible to deliver results for such cases in near real-time in case of the event leveraging LEXIS platform.

In the prototype form presented in this paper, this workflow is already a powerful tool for emergency teams

and urgent responders in case of such a disaster happenstance. Being a major improvement over the state of the art Satellite Base Mapping for such events, the ultimate goal outlined in the future work would further improve the efficiency of the computing algorithms, develop sensible results aggregation for the TsunAWI ensemble runs and add path for the precise inundation computation as shown in Figure 6.

This final workflow would provide both a real-time early warning part, as described in this paper on a coarse-grain evaluation, and a second, slowest but closer to real-life loss and inundation assessment a few dozen minutes after the end of the earthquake. The the second, fine-grained tsunami simulation would be run in about a dozen minutes after the acquisition of all the parameters, depending on the size of the simulated area. The final refined simulation over the SEM can be received in dozens of minutes to a few hours at most. The impact of such a technology would be major.

This does not dwarf the fact that the fully implemented workflow as demonstrated in the LEXIS project on the platform's infrastructure even without the future development is a major improvement over the state of the art in the domain.

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Notes

1. Consistency or self-consistency for a SDF/CSDF graph means there exists at least one vector of number of firing of the processes (i.e. components or microservices when using cloud terminology) of the data-flow that allows for the activation of all processes while ensuring all the necessary data-tokens has been produced and consumed over the activation of the processes. This also means that the absence of deadlock. The determinism within a MoC means that any same inputs is guaranteed to give the same outputs independently of discrepancies in scheduling, communication timing, process fine-grain firing, or allocation of computing resources. Therefore, it is a desirable property of a MoC since that it allows to replay error scenarios, debug a given workflow by replaying an error scenario previously encountered, or more powerfully enabling validation of the whole system by validating and testing all its parts. Another important property of a MoC is its ability of abstracting the design and implementation of a given complex specification in a easy to grasp and verify model. Below, we explain the bases of the MoC we utilize and showcase how it abstract the design in a very simple way is verifiable easily at design time, that the data-flow can run in finite memory without deadlocking, and overall that the specification of the parallelism in the data-flow is correctly done.
2. Which means the denoted communication channels are connected from the output port of the considered origin process to the input port of the destination process.
3. It is also possible to authorize more than one token at a time on a given output channel, but as it is a rare use-case, we chose to not discuss it further for the sake of clarity.
4. More on what hypotheses are required to fulfill such requirement later.
5. https://irods.org/.
6. At 99.99% availability any server should have less than 1 hour of downtime annually. Therefore at 3 simultaneous unavailable servers, the probability would be of the order of 10^{-12} which can be considered safe, unless there exist common root cause (e.g. shared power lines but the servers must be selected adequately).
7. https://www.it4i.cz/en.
8. https://www.lrz.de/.

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